

Endong Dai

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PROFILE

Hardware engineer with 5 years in digital and mixed-signal systems, spanning RTL design and verification, FPGA prototyping, post-silicon SoC bring-up, and PCB design. Drove an AI accelerator from tape-out through full silicon validation on custom carrier platforms.

EDUCATION

Duke University	M.S. in ECE -Hardware Track (GPA 3.8/4.0 First Year)	Aug 2025 – May 2027
University of Nottingham	B.Eng. in EEE, First Class (GPA 3.8/4.0)	Sep 2021 – Jul 2025

TECHNICAL SKILLS

- **IC & FPGA Design:** Verilog, Vivado, Quartus Prime, ModelSim, FPGA (Xilinx ZCU102), Cadence Virtuoso
- **PCB & Simulation:** Altium Designer, KiCad, LTspice, PLECS, MATLAB/Simulink, Soldering
- **Programming & Embedded:** C/C++, Assembly(PIC16, MIPS), STM32CubeIDE, Arduino, Raspberry Pi, ROS
- **Tools & Platforms:** Claude Code, Linux(Ubuntu), Git

PROJECT EXPERIENCE

LLM Accelerator RTL Design – Matrix Multiplication Unit (MMU) Jan 2026 – Jun 2026

RTL Design Engineer – Duke Center of Computational Evolutionary Intelligence (CEI)

- Architected the Matrix Multiplication Unit (MMU), an LLM accelerator's central GEMM/GEMV engine; a K-means codebook quantization recasts memory-bound decode GEMV as dense GEMM for a **10× decode speedup**.
- Implemented two native operating modes on a shared PE array and control path: full-precision prefill GEMM and codebook-indexed quantized GEMV for optimized decode.
- Designed a **32×32 INT8 systolic PE array (1,024 MACs)** with AXI-Stream ready/valid dataflow delivering **~1.0 TOPS**, verified against software golden models and synthesized in a **TSMC 16 nm** flow.

Diffusion Chip – FPGA & SoC Bring-Up

Aug 2025 – Jun 2026

System Bring-up / RTL Design Engineer – Duke CEI

- Drove backend productization and bring-up of the diffusion chip (TSMC **16nm CPGA-208**): CPU/NPU, 1 MB GLB SRAM, GPIO, and **dual 32-bit FX3** on a shared AXI fabric (**INT8 1.024 TOPS**, FP16 0.256 TOPS).
- Deployed FPGA-based validation on AMD **ZCU102** to verify PC-to-chip communication and control functionality through FX3, UART, JTAG, and GPIO interfaces during system bring-up.
- Performed post-silicon SoC validation on the taped-out chip, bringing up **FX3 high-speed TX/RX** links and running **NPU and general-purpose operators** on real silicon against software golden models.

CPGA-208 Diffusion Chip Carrier PCB Design

Feb 2026 – Jun 2026

PCB Project Engineer – Duke CEI

- Led chip bonding and **PCB** design of a **CPGA-208** carrier board for the Fornax diffusion chip on a 6-layer **Altium Designer** platform, including **length matching** for **dual 32-bit** high-speed FX interfaces.
- Designed **multi-domain power** with LDO-based regulation, level shifting and power filtering networks.

Cadence Virtuoso IC Design (TSMC 65nm) – 16-bit ALU & PLL FM Receiver

Sep 2025 – Apr 2026

Digital / Analog-Mixed-Signal IC Design Engineer (Course Project) – Duke University

- Designed schematic and full-custom layout of an **IEEE-754 half-precision ALU** (16 MHz, <18 mW) and a PLL-based multi-channel FM receiver (VCO, phase detector, charge pump).

TEACHING EXPERIENCE

Teaching Assistant – Duke University

Summer – Fall 2026

Department of Electrical & Computer Engineering

- Led an RTL design workshop (Summer 2026), guiding students through Verilog modeling, simulation.
- Course TA for *ECE550 Fundamentals of Computer Systems and Engineering* (Fall 2026): held office hours, graded assignments, and mentored students on RTL design and verification.